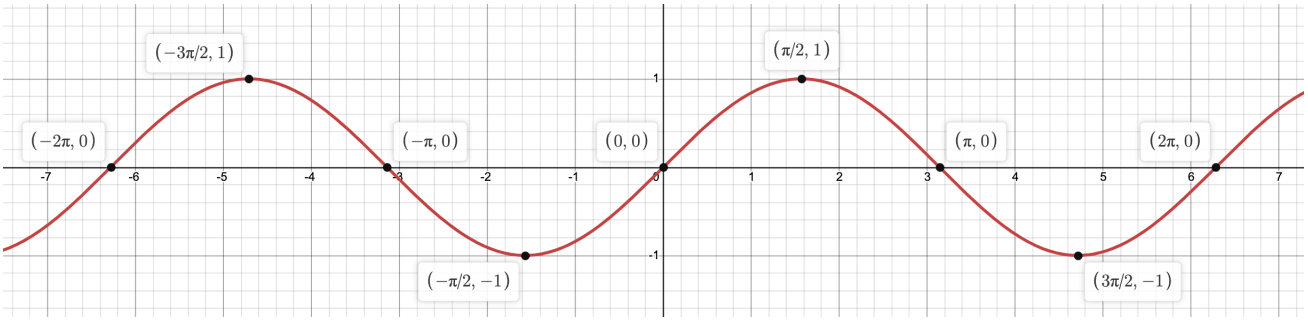
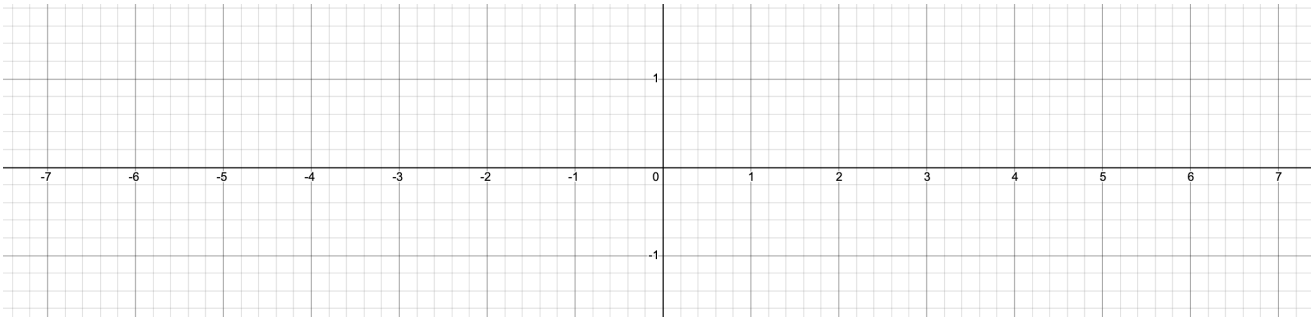
1. Consider the graph of $f(x) = \sin(x)$. Sketch the graph of $f'(x)$ on the axis provided below. Using your graph, can you make a conjecture for the function $\frac{d}{dx}\sin(x)$?

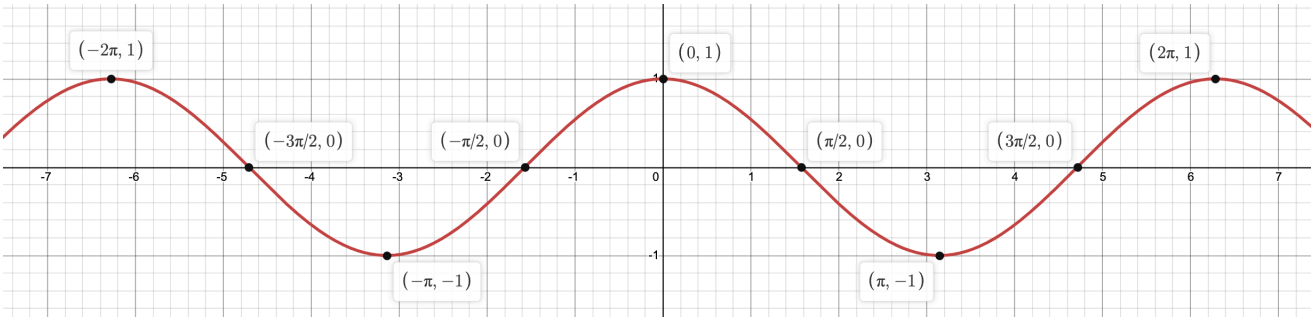


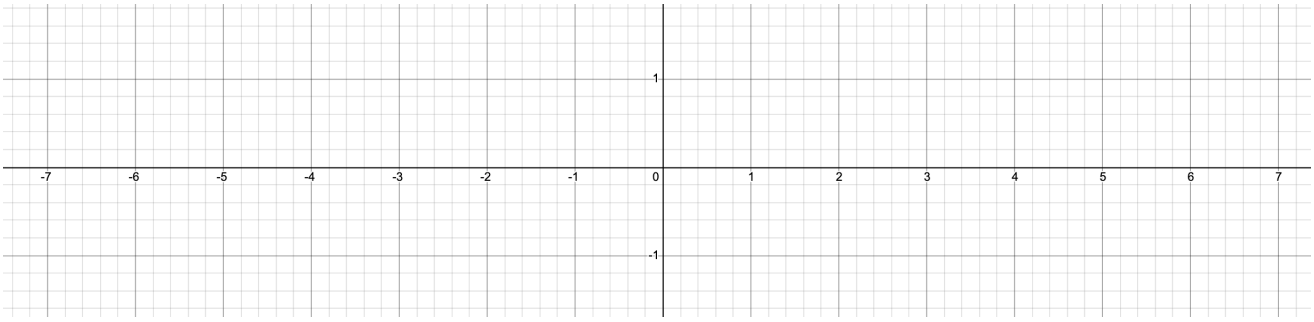


2. Prove your conjecture by computing $\frac{d}{dx}\sin(x)$ using the definition of the derivative.
Hint: Recall the trig identity $\sin(a+b) = \sin(a)\cos(b) + \cos(a)\sin(b)$.

3. Differentiate $y = x^2 + 2\sin(x)$.

4. Consider the graph of $f(x) = \cos(x)$. Sketch the graph of $f'(x)$ on the axis provided below. Using your graph, can you make a conjecture for the function $\frac{d}{dx}\cos(x)$?





5. Prove your conjecture by computing $\frac{d}{dx}\cos(x)$ using the definition of the derivative.
Hint: Recall the trig identity $\cos(a+b) = \cos(a)\cos(b) - \sin(a)\sin(b)$.

6. Find the equation of the tangent line to the curve $y = x + \cos(x)$ at the point $(\frac{\pi}{4}, \frac{\pi}{4} + \frac{1}{\sqrt{2}})$.

Consider $f(x) = a^x$ for $a > 0$.

7. Use the definition of the derivative to write an expression for $f'(0)$.

8. Use the definition of the derivative to write an expression for $f'(x)$. Simplify this expression as much as you can.

9. From the above, we see that $f'(x)$ exists for all x if $f'(0)$ exists. Does $f'(0)$ exist? To investigate, access the Desmos animation here `https://www.desmos.com/calculator/f7hbdcbui5`. Use the slider to vary a.

   (a) Does it look like $f'(0)$ exists for $a = 2$? If so, give an approximate value.

   (b) Does it look like $f'(0)$ exists for $a = 3$? If so, give an approximate value.

   (c) Do you think there is a value of a for which $f'(0) = 1$? Give an upper and lower bound for a.

   (d) Can you think of any reason why such an *a must* exist?

10. As we guessed, there turns out to be a unique number, which we denote by e, such that $\lim_{h\to 0} \frac{e^h-1}{h} = 1$.

    (a) Calculate $\frac{d}{dx}e^x$.

    (b) If c is any constant, calculate $\frac{d}{dx}(ce^x)$.

11. Differentiate $y = \sqrt{3}\sin(x) + 2\pi\cos(x) - 3e^x$.